TERMINAL EXCAMS - 2020.

CHEMISIRY MARKING SCHEME.

FORM THREE.

SECTION:A.

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
E	E	B	D	D	B	B	C	C	D

1@ = 10 marks

2.

i	ii	iii	iv	v
H	D	B	F	C

1@ = 5 marks

SECTION : B

3a) (i) -> Hydrogen peroxide - H_2O_2

-> Potassium chlorate - $KClO_3$

(ii) $\Rightarrow 2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$

$\rightarrow 2KClO_3 \xrightarrow{MnO_2} 2KCl + 3O_2$

(iii) Downward displacement of water

5) (i) Nitric acid not used in the laboratory preparation of hydrogen because it is a strong oxidizing agent; oxidises Hydrogen into water.

(ii) When a burning wooden splint is lowered in the a test tube containing hydrogen gas, it explodes with a "pop" sound. 1@ = 7 marks

4. a) (i) Group II Element ... 1 marks

(ii) Chemical equation

$Mg_{(s)} + 2H_2O_{(l)} \longrightarrow Mg(OH)_{2(aq)} + H_{2(l)}$ ... 1 marks

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(b) (i) 1. It saves time and space in writing

2. In order to prepare a definite amount of the product, the amount of the reactant to be used can be calculated accurately.

3. A single arrow b/n the reactants and products tells us that the reaction goes to completion while a double arrow shows that it is a reversible reaction.

03 marks

ii) $\left(\frac{X}{100} \times 220\right) + \left(\frac{X}{100} \times 224\right) = 223$

$100 \times \left(\frac{220X + 224X}{100}\right) = 223 \times 100$

$\frac{444x}{444} = \frac{22300}{444}$

$x = 50.23\%$

$\therefore \frac{220}{223} X$ is 50.23%

$\frac{224}{223} X$ is 50.23%.

02 marks 02

5. a) Ionic equation is chemical equation in which compounds in aqueous solution or in molten state are written as dissociated ions while Molecular equation is the equation which shows the actual number of each different atom in a molecule.

02 marks

b) (i) $3KOH_{(aq)} + FeCl_{3(s)} \longrightarrow 3KCl_{(aq)} + Fe(OH)_{3(s)}$

$3K^+_{(aq)} + 3OH^-_{(aq)} + Fe^{3+}_{(aq)} + 3Cl^-_{(aq)} \longrightarrow 3K^+_{(aq)} + 3Cl^-_{(aq)} + Fe(OH)_{3(s)}$

02 marks

(ii) $3OH^-_{(aq)} + Fe^{3+}_{(aq)} \longrightarrow Fe(OH)_{3(s)}$. 02 marks

(iii) Double displacement / precipitation reaction.

01 marks

6 (a) Because of metallic properties when you move across the period is decreases and show ~~metallic~~ non-metallic properties. 2 marks

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(b) Cations slightly smaller than their neutral atoms because of loosing number of electrons and Anions are slightly larger than their neutral atoms because of gaining number of electrons. 3 marks

(c) Water is a universal solvent because water dissolves more substances than any other liquid. 2 marks

7. (a) (i) Standard solution: Is the solution of known concentration. 01 maks

(ii) Data

Molarity = ?

Percentage purity = 98% 01/2 marks

Density / Assay = 1.63 g/cm^3

formula

$$\text{Molarity} = \frac{\%\ \text{purity} \times \text{Assay} \times 1000}{\text{Molar mass}}$$ 01/2 marks

Molar mass = H_2SO_4 = 98 g/mol.

$$= \frac{98/100 \times 1.63\ g/dm^3 \times 1000}{98\ g/mol.}$$ 01/2 marks

Molarity = 16.3 M. 01/2 marks

(b) Data

Mass of hydrated = 2.62g in 1L

M_b = 25cm^3

V_a = 23cm^3 01/2 maks

7
8
8
7

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Conc. of Acid = 1.98g in Litre

Relative atomic mass = ?

Chemical reaction

$Na_2CO_3 + 2HCl \Rightarrow 2NaCl + CO_2 + H_2O$ 0 1 marks

Molarity = $\frac{\text{Concentration}}{\text{Molar Mass}}$ 0 1/2 mark

Mola mass Na_2CO_3 = 106g/mol.

$= \frac{2.62\ g/dm^3}{106\ g/mol.}$

$= 0.025M$ 0 1/2 marks

formula

$\frac{M_a V_a}{M_b V_b} = \frac{n_a}{n_b}$ 0 1/2 marks

$M_a = \frac{M_b V_b n_a}{V_a n_b} = \frac{0.025 \times 25 \times 2}{23 \times 1} = 0.05M$.

Molarity = $\frac{\text{concentration}}{\text{Molar mass}}$ 0 1/2 marks

Molar mass = $\frac{\text{Conc.}}{\text{Molarity}}$

$= \frac{1.98\ g/dm^3}{0.05M}$

$= 39.6$

HQ $= 39.6 - 1$

Q $= 38.6$ 0 1/2 marks.

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8. (a) $CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

(b) $KOH(aq) + HNO_3(aq) \longrightarrow KNO_3(aq) + H_2O(l)$

(c) ~~$NaNO_3$~~ + $AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + NaNO_3(aq)$.

(d) $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

(e) $Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$.

2@ = 10

9a)

Element	C	H
%	85.7	100 − 85.7 ½
	$\frac{85.7}{12}$	$\frac{14.3}{1}$ ½
	$\frac{7.14}{7.14}$	$\frac{14.3}{7.14}$ ½
	1	2 ½

½ @ 02 marks

∴ Empical formula = CH_2. 01 marks

b $(E.F)_n$ = Molecular weight

$(CH_2)_n = 56$

$(12+2)n = 56$

$14n = 56$ 01

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$$\frac{14n}{14} = \frac{56}{14}$$

$$n = 4$$

$$(CH_2)_4 = C_4H_8$$

∴ Molecula formula = C_4H_8 01 marks

(c) chemical reaction

$$C_4H_8 + O_2 \longrightarrow CO_2 + H_2O$$

$$C_4H_8(aq) + 6O_2(g) \longrightarrow 4CO_2(g) + 4H_2O(l)$$ 00 1/2 marks

When compound burn in Air produce Co_2 gas which turn lime water into milk and water. 01/2 marks

10. a)(i) Washing Soda is added to permanent hardness of water to form a precipitated compound of Calcium carbonate, then filtration taking place to obtain soft water 01 marks

$$CaSO_4(aq) + Na_2CO_3(aq) \longrightarrow CaCO_3(S) + Na_2SO_4(aq)$$ 01 marks

ii) Temporary hard water removed by Boiling promote the formation of carbonate from the hydrogen carbonate and precipitate the carbonate out of the solution. 1 marks

$$Ca(HCO_3)_2(aq) \longrightarrow CaCO_3(S) + CO_2(g) + H_2O(l)$$ 01 marks

b) (i) Due to the formation of Calcium Carbonate

$$CaCO_3 \rightarrow Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(S) + H_2O(l)$$ 01 marks

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10 (b)(ii) Ion exchanger remove all calcium and Magnesium ions in the water while through boiling Calcium and Magnesium ion can not be removed.
01 marks

11. (a) (i) $P_{(s)} + 3O_2 \longrightarrow 2P_2O_3$

(ii) $K_2CO_{3(s)} + 2HCL_{(g)} \longrightarrow 2KCL_{(aq)} + H_2O_{(l)} + CO_{2(g)}$

(iii) $Fe_{(s)} + CuSO_{4(aq)} \longrightarrow Cu_{(s)} + FeSO_{4(aq)}$
01 @ 03 marks

(b) (i) Used to control soil pH
(ii) Used as ant-acids
(iii) Used as inorganic fertilizers
(iv) Used to alleviate health disorders
(v) Used as fungicide.
vi) Used for seasoning and preventing food.
(vii) Used to Manufacture dry batteries.
only 4 points = 04 marks

12. a) (i) Molar mass: Is the mass of one mole of Substance. 01 marks

(ii) Molar Volume: Is the volume of one mole of any gas at standard temperature and pressure, which is $22.4 dm^3$
01 marks

(b) (i) $Al_2(SO_4)_3$

Solution.

Molar mass of:

Al = 27 g/mol.
S = 32 g/mol
O = 16 g/mol

$= (27 \times 2) + (32 \times 3) + (16 \times 4) \times 3$
$= 54 + 96 + 192$
$= 342 g/mol$. ∴ Molar Mass of $Al_2(SO_4)_3 = 342 g/mol$.
01 marks

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12(b) ii) $Na_2CO_3 \cdot 10H_2O$

$= (Na \times 2) + C + (O \times 3) + 10(H_2O)$

$= (23 \times 2) + 12 + (16 \times 3) + (10 \times 18) =$

$= 46 + 12 + 48 + 180$

$= 286\ g/mol.$ 01 marks

∴ Molar mass of $Na_2CO_3 \cdot H_2O = 286\ g/mol.$

c (i) Data

Mass = 5.85g

Volume = $100cm^3$ ($0.1dm^3$)

Molarity = ?

Molar mass (Nacl) = 23 + 35.5 = 58.5 g/mol.

Formula

$$\text{Molarity} = \frac{\text{mass}}{\text{Molar mass} \times \text{volume}}$$

$$= \frac{5.85\ g}{58.5\ g/mol \times 0.1\ dm^3}$$

Molarity = 1 M 01 marks

(ii) Data given

Mole = 0.2 mol

Volume = $11.2dm^3$

Molarity = ?

Formula

$$\text{Molarity} = \frac{\text{Number of mole}}{\text{volume}}$$

$$\text{Molarity} = \frac{0.2\ mol}{11.2\ dm^3}$$

$$= 0.017\ M$$

∴ Molarity of $FeCl_3$ = 0.017 M. 01 marks

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13. - Introduction - Any relevant introduction 1½ marks

- Chemistry as other science subject involves systematic ways of discovery of knowledge and Solving of problems. Discovery of knowledge and Solving problems involves very careful investigation Such as

(i) Problem Identification
(ii) Formulation of hypothesis
(iii) Experiment and Observation
(iv) Data collection and analysis
(v) Data interpretation 2@ = 12
(vi) Drawing a conclusion.

- Conclusion {any relevant conclusion} · 1½ marks

14. a)(i) Mole : Is the amount of the substance which contains as many particulate entities as there are carbon atom in 12g of Carbon-12. 02marks

(ii) Molar Solution : Is the solution which contain 1mole of the solute dissolved to make 1 litre of the solution. 02marks

b) (i) Data given

Volume = $11.2 dm^3$
Molar volume = $22.4 dm^3$ 0½ marks
Number of mole = ?
Formula
Number of mole(n) = $\frac{\text{Volume}}{\text{Molar volume}}$ 0½ marks

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4.b i)

$$= \frac{11.2\,dm^3}{22.4\,dm^3}$$

$$= 0.5\,mol.$$

Number of moles = 0.5 mol 01 marks

(ii) Data given

Mass = 20g

Molar Mass of Al_2SO_4 = (27x2) + 32 + (16x4) =
54 + 32 + 64
150 g/mol. 01/2 marks

Avogadro's constant = 6.02×10^{23} 01/2 marks

Number of particulate entities(N) = ?

$Al_2SO_4 = 2Al^{3+} + SO_4^{2-}$ 01 marks

= 3 ions

$$n = \frac{mass}{Molar\ mass}$$ 01/2 marks

$$n = \frac{20\,g}{150\,g/mol.}$$

$n = 0.133\,mol \times 3\,ions$

$= 0.399\,mols.$

$N = n L_A$

$N = 0.399\,mols \times 6.02 \times 10^{23}$

$= 2.4 \times 10^{23}$ ions. 01

c Data given

Mass of $CaCO_3$ = 5.6g

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Volume of $CO_{2(g)}$ = ?

Molar Volume = ? 00 1/2 marks

Molar mass of $CaCO_3$ = 100g.

Solution

$$CaCO_{3(s)} \longrightarrow CaO_{(s)} + CO_{2(g)}$$ 01 marks

$$CaCO_{3(s)} \equiv CO_{2(g)}$$

$$n = \frac{\text{mass}}{\text{Molar mass}}$$ 01 marks

$$n = \frac{5.6\ g}{100\ g/mol}$$

Number of mole = 0.056 mol. 01 marks

$$0.056\ mols = \frac{\text{Volume}}{22.4\ dm^3}$$ 01 marks

$$\text{Volume} = 0.056\ moles \times 22.4\ dm^3$$

$$= 1.25\ dm^3.$$ 01 marks

OR.

$CaCO_3 \equiv CO_2$ at S.T.P

$100g \equiv 22.4\ dm^3$

$5.6g \equiv ?$

$$= \frac{5.6\ g \times 22.4\ dm^3}{100g} = 1.25\ dm^3$$

∴ Volume of CO_2 is $1.25\ dm^3$

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